

H-Bridge Based Motor Driver Design

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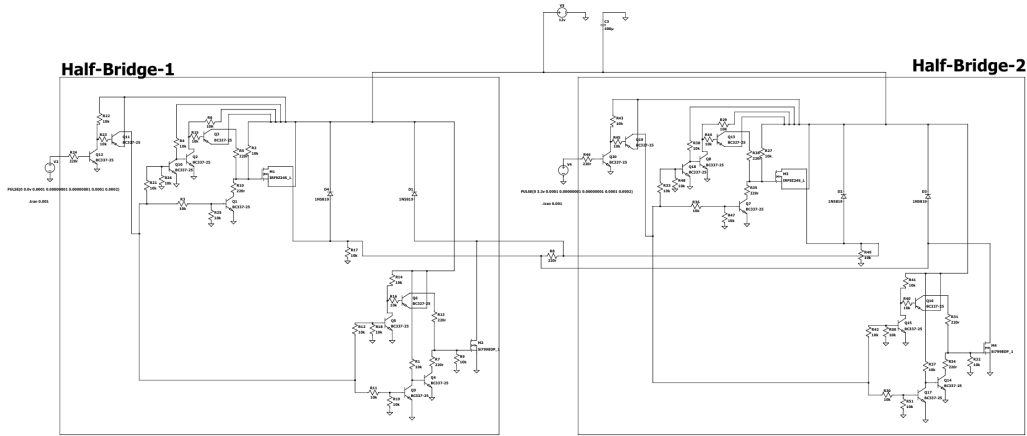
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Bölüm 1

H-Bridge Based Motor Driver Circuit

This study is a continuously developing *living document* that examines the design processes of discrete power electronics topologies. The current report includes the v1.0 stability tests, simulation outputs of the first phase of the project, and the initial thermal/characteristic laboratory findings obtained from the physical prototype.

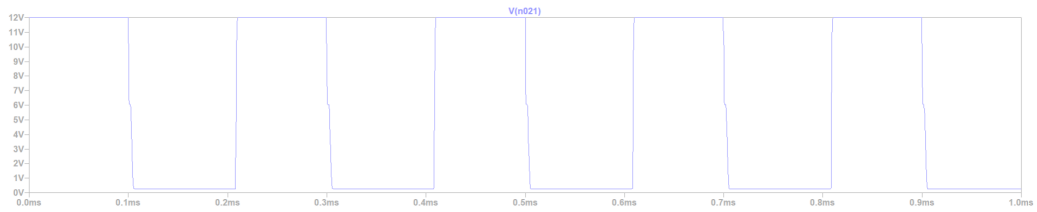


Şekil 1.1: H-Bridge based motor driver circuit schematic driving a 220Ω resistive load

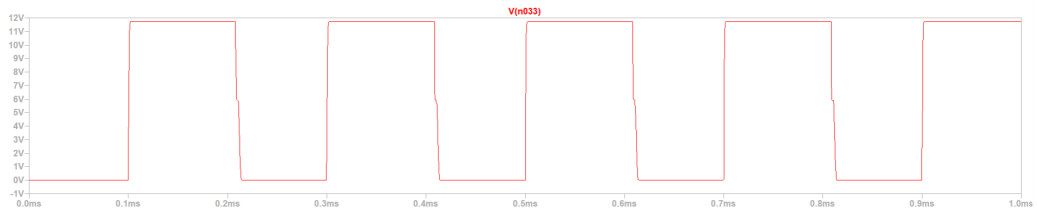
Bölüm 2

Simulation Data for 5 kHz Triggering Signal

2.1 Triggering Signals

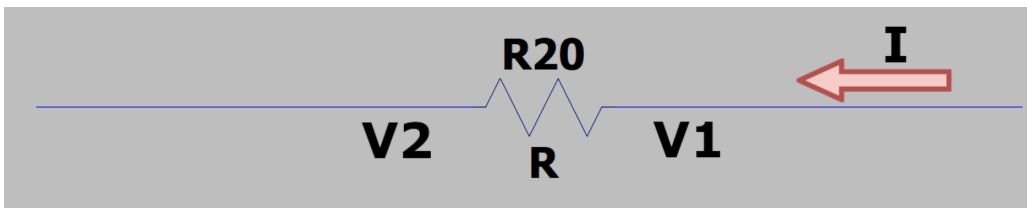


Şekil 2.1: PMOS Gate Terminal Voltage Waveform



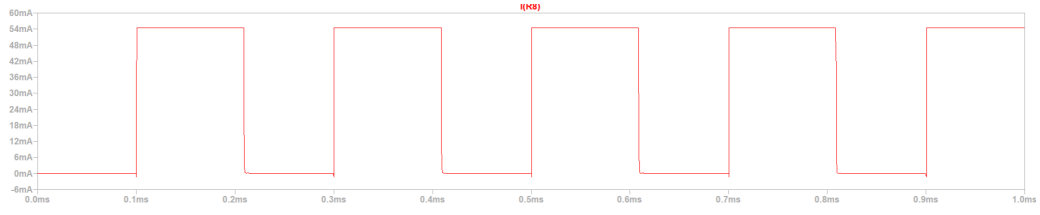
Şekil 2.2: NMOS Gate Terminal Voltage Waveform

2.2 Driven Load: Resistive

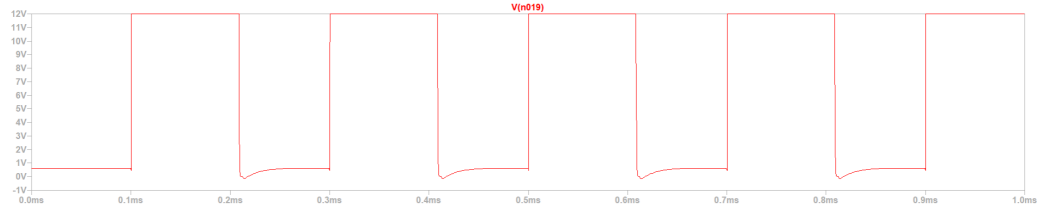


Şekil 2.3: Schematic representation of the measured variables to be plotted

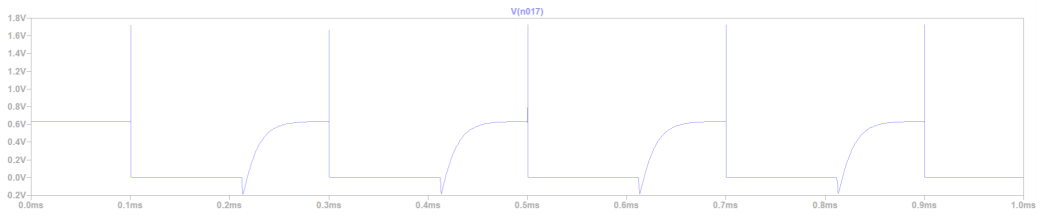
2.2.1 220 Ω Resistor



Şekil 2.4: Current waveform for the 220 Ω resistor

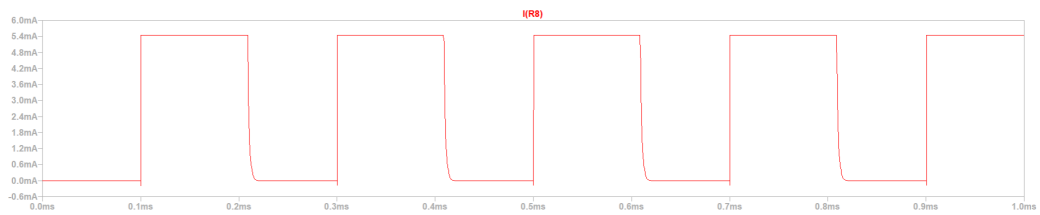


Şekil 2.5: V1 voltage waveform for the 220 Ω resistor

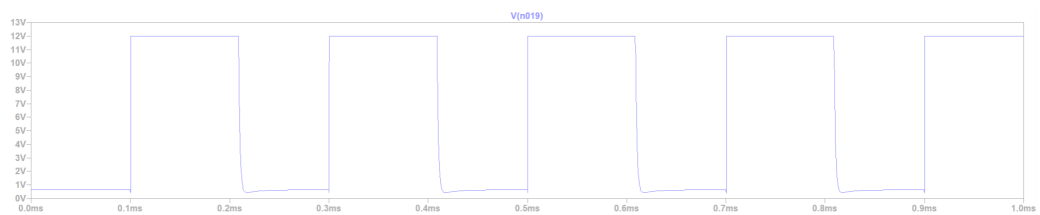


Şekil 2.6: V2 voltage waveform for the 220 Ω resistor

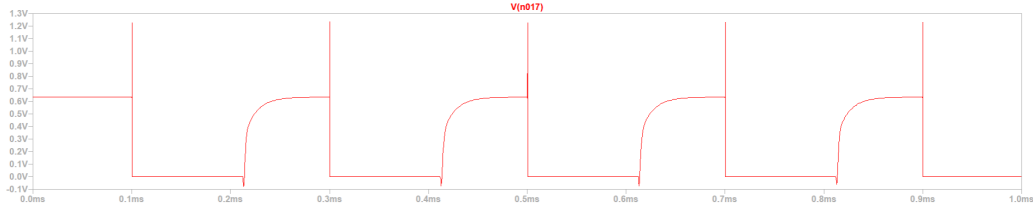
2.2.2 2.2 k Ω Resistor



Şekil 2.7: Current waveform for the 2.2 k Ω resistor

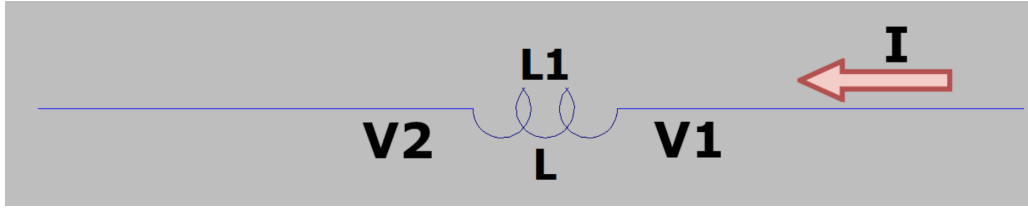


Şekil 2.8: V1 voltage waveform for the 2.2 k Ω resistor



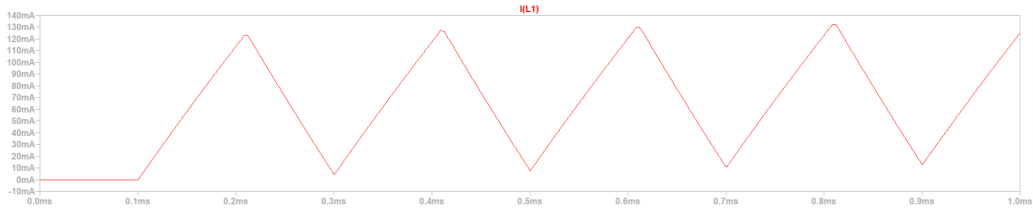
Şekil 2.9: V2 voltage waveform for the 2.2 kΩ resistor

2.3 Driven Load: Inductive

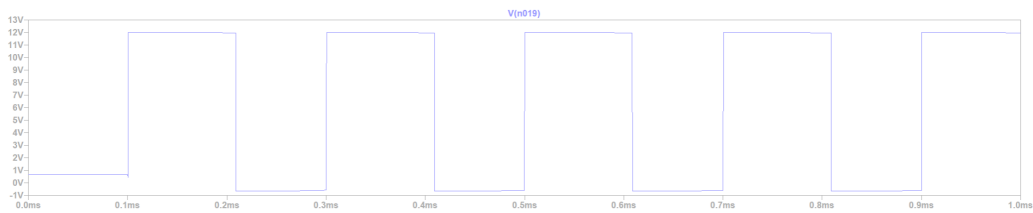


Şekil 2.10: Schematic representation of the measured variables to be plotted

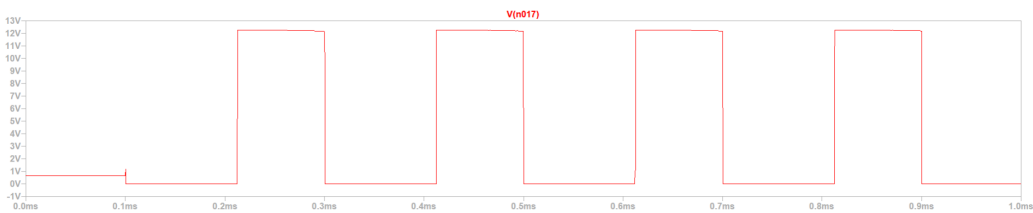
2.3.1 Coil (10 mH Inductance and 10 Ω Internal Resistance)



Şekil 2.11: Current waveform for the 10 mH – 10 Ω coil



Şekil 2.12: V1 voltage waveform for the 10 mH – 10 Ω coil



Şekil 2.13: V2 voltage waveform for the 10 mH – 10 Ω coil